REPORT

Reviewer report: urllib3/urllib3#5010: Add

retry_after_max_strict to raise on excessive Retry-After

Run urllib3-5010, tested 2026-11-13 11:41 IST by ops/daily_run.py (route source-pair). Proof-of-Change commit

ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-13 12:33 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

NEEDS ATTENTION

Recommended action: Reject Confidence: High

The run is trustworthy and found a difference a person should look at. Every check passed and every claim cites

a log line.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: Add retry_after_max_strict to raise on excessive Retry-After (urllib3/urllib3#5010).

• What we observed: 11 new test(s) fail on the PR; 366 test(s) ran on main, 391 on the PR; 14 new test(s) pass,

119 unchanged (7 of them skipped on both sides); 254 test failure(s) pre-existing and identical on both sides.

• What you should do: Do not approve as is. Read 'What changed in behavior' and decide with the PR author.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 6.4 GB (of 15.6 GB).

• Setup check: free disk 91.6 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: urllib3 python setup install instructions https://github.com/urllib3/urllib3 site:github.com/urllib3/urllib3

• Query: urllib3 python documentation install https://github.com/urllib3/urllib3 site:github.com/urllib3/urllib3

• Query: urllib3 python install requirements https://pypi.org/project/urllib3/ https://github.com/urllib3/urllib3

site:pypi.org

• Source: urllib3 · PyPI — https://pypi.org/project/urllib3/

• Source: GitHub - urllib3/urllib3: urllib3 is a user-friendly HTTP client library for Python · GitHub —

https://github.com/urllib3/urllib3

• Source: urllib3 installation requires setuptools >= 36.2.5 · Issue #1753 · urllib3/urllib3 · GitHub —

https://github.com/urllib3/urllib3/issues/1753

• Source: urllib3/CHANGES.rst at main · urllib3/urllib3 · GitHub —

https://github.com/urllib3/urllib3/blob/main/CHANGES.rst

• Source: Installing urllib3 · Issue #829 · urllib3/urllib3 · GitHub — https://github.com/urllib3/urllib3/issues/829

• Setup plan: the same packages on main and pr.

• Setup plan: install trustme>=1.2.1 (listed in the test extra) on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.10

• Installed trustme 1.2.1 from pyproject (listed in the test extra) on main and pr.

• Before rerun: needs_attention: 11 new test(s) fail on the PR; 366 test(s) ran on main, 391 on the PR; 14 new

test(s) pass, 119 unchanged (7 of them skipped on both sides); 254 test failure(s) pre-existing and identical on

both sides

• After rerun: not rerun. The collection import was a setup gap, but it was not repaired, so the recorded verdict

was kept.

Setup: pytest plugin asyncio was turned off. Pytest stopped at startup the same way on both sides, and the

traceback pointed into that third-party plugin (AttributeError: module 'pytest' has no attribute

'pyodide_options_stack'). The tests were run again with the plugin turned off. This does not change how the

verdict is chosen. Detection fired. Pytest stopped at startup the same way on both sides. Plugin chosen: asyncio.

The installed pytest plugin list was read. Retry flags: -p no:asyncio. Outcome: worked.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| C1 | Add retry_after_max_strict to raise on excessive Retry-After (PR title) | not testable | claims are read out of

the PR description only by model-written probes (--model-provider, see Probes); the tests below check behavior |

pytest-pr-1.log:34668 «132 failed, 126 passed, 7 skipped, 147 warnings, 265 errors in 170.87s (0:02:50)»

| C2 | Existing behavior is kept: tests that pass on main still pass (implied by every PR) | false | 11 test(s) fail on

the PR, e.g.

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-False] |

pytest-pr-1.log:34422 «ERROR test/with_dummyserver/test_poolmanager.py::TestPoolManager::test_redirect_r

etry_after_max_strict[http-False]»

| - | Change not in the PR description (this run) | unmentioned change | 11 test(s) are new in the PR and fail, e.g.

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-False] |

pytest-pr-1.log:34422 «ERROR test/with_dummyserver/test_poolmanager.py::TestPoolManager::test_redirect_r

etry_after_max_strict[http-False]»

Citation check (PASS): 3 citation(s), every one re-read from its log file and matched.

What changed in behavior

| What | Count | Examples

| Tests that actually ran (main / PR, fewest of the runs) | 366 / 391 | | Tests unchanged | 373 | | New tests that pass | 14 | | Tests that pass on main but fail on the PR | 0 | | Tests that are new in the PR and fail | 11 |

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-False],

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-True],

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[https-False],

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[https-True], te

st.with_dummyserver.test_socketlevel.TestRetryAfterMaxStrict::test_connection_pool_raises_without_retrying[F

alse-302 Found]

| Tests that ran on main but are missing on the PR side | 0 | | Tests that changed result in another way (for example passed -> skipped) | 0 | | Tests that gave different results between the 3 runs of

the same side (flaky) | 0 | | Tests that fail on main and pass on the PR | 0 | | Tests that are new or changed but did not run because an optional module is missing | 0 | | Tests that were re-parametrized (same test, new parameter ids; not counted as removed) | 0 | | Test modules that fail to load on the PR side only (their tests no longer run) | 0 | | Test modules that fail to load on main only (no baseline for their tests) | 0 | | Test modules that fail to load on both sides (their tests did not run) | 0 | | Tests that fail identically on both sides (already broken on main) | 254 | test.with_dummyserver.test_poolmanag

er.TestPoolManager::test_303_redirect_makes_request_lose_body[http], test.with_dummyserver.test_poolmana

ger.TestPoolManager::test_303_redirect_makes_request_lose_body[https], test.with_dummyserver.test_poolma

nager.TestPoolManager::test_303_redirect_makes_request_lose_body_framing[http-header], test.with_dummys

erver.test_poolmanager.TestPoolManager::test_303_redirect_makes_request_lose_body_framing[http-kwarg], t

est.with_dummyserver.test_poolmanager.TestPoolManager::test_303_redirect_makes_request_lose_body_fram

ing[https-header]

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-False]

fails with: failed on setup with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[http-True] fails

with: failed on setup with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

test.with_dummyserver.test_poolmanager.TestPoolManager::test_redirect_retry_after_max_strict[https-False]

fails with: failed on setup with "OSError('failed to bind socket') [single exception in ExceptionGroup]"

Probes

Model-written probes: model provider: nebius (Token Factory, live).

No probe ran for this PR (no fixed probe in probes/<owner>/<repo>/<pr>.py and no model-written or differential

probe), so only the tests were compared.

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (177.4 s run).

Coverage gate (python, confidence normal): 14 of 14 changed executable line(s) ran (100%).

| Changed file | Changed executable lines | Of those, ran

| src/urllib3/exceptions.py | 8 | 8

| src/urllib3/util/retry.py | 6 | 6

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested c0d6fccfa8c4 of the PR against main (source-pair route).

• 3 alternating runs per side: main 3, PR 3 of 3.

• pytest exit codes explained: pytest exit 1; 254 failure(s)/error(s) pre-existing and identical on both sides.

• Changed code was exercised (coverage gate): 14 of 14 changed executable line(s) ran (100%).

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every claim cites a real log line: 3 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

| Issue | Status | Detail | How to fix

| Behavior difference found | finding | 11 new test(s) fail on the PR; 366 test(s) ran on main, 391 on the PR; 14

new test(s) pass, 119 unchanged (7 of them skipped on both sides); 254 test failure(s) pre-existing and identical

on both sides | This is about the PR, not the run: read the failing tests and their log lines, then decide.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | PASS | main 3, PR 3 of 3

| pytest exit codes explained | PASS | pytest exit 1; 254 failure(s)/error(s) pre-existing and identical on both sides

| Changed code was exercised (coverage gate) | PASS | 14 of 14 changed executable line(s) ran (100%)

| Probes ran on both sides | SKIP | no probe for this PR

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

| Nothing posted upstream | PASS | GitHub was only read

| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

| Every claim cites a real log line | PASS | 3 citation(s), every one re-read from its log file and matched

Checklist

• I read the verdict and the plain-English summary.

• Every 'false' or 'unmentioned change' row has a log line I can open.

• Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).

• I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide REPORT=poc-main/reports/daily/2026-11-13/urllib3-5010 DECISION=approve (or reject, approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/urllib3/urllib3/pull/5010

Evidence links

• Pull request urllib3/urllib3#5010

• record.json

• pr.diff

• setup-main.log

• setup-pr.log

• git.log

• pytest-main-1.log

• pytest-pr-1.log

• pytest-main-2.log

• pytest-pr-2.log

• pytest-main-3.log

• pytest-pr-3.log

• junit-main-1.xml

• junit-pr-1.xml

• coverage-pytest.log

• coverage-pr.json

• main run 1 result: pytest-main-1.log:33171 «125 failed, 112 passed, 7 skipped, 140 warnings, 254 errors in 165.89s (0:02:45)»

• pr run 1 result: pytest-pr-1.log:34668 «132 failed, 126 passed, 7 skipped, 147 warnings, 265 errors in 170.87s

(0:02:50)»

Glossary

| Term | Meaning

| main | The library as it is today, without the PR.

| PR | Pull request: a proposed change to the library.

| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.

| flaky | A test that gives different results when nothing changed; it is ignored as evidence.

| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-13 12:33 IST